Answer Key

Area

Area is how many squares cover a shape.

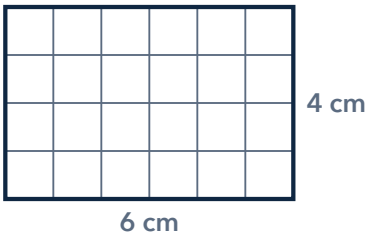
Cover it with squares, then count

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15

Area = 15 square units

Each square is 1 square unit.

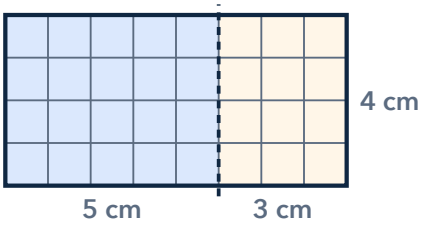
Rows × columns: no need to count one by one



4 × 6 = 24

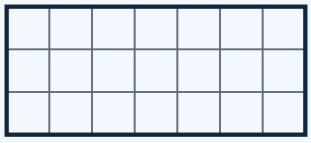
4 rows of 6 squares: 24 square cm.
Multiply the side lengths.

Break it apart: 4 × 8 is 4 × 5 and 4 × 3



4 × 5 = 20
4 × 3 = 12
20 + 12 = 32 square cm

Your turn!



3 × 7 = 21 square units

For teachers

Goal: area as unit squares that cover a shape, then rows × columns (3.MD.5–7). Say "How many rows? How many in each row?" Area is inside; the edge is perimeter (next packet). Link breaking apart to packet 34.

Answer Key

Area: Practice

Count the squares, or multiply rows × columns.

Part 1 Count the squares Each square is 1 square unit.

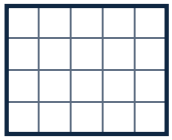
1) Area = 12 square units



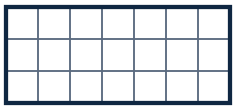
2) Area = 14 square units



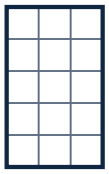
3) Area = 20 square units



4) Area = 21 square units



5) Area = 15 square units

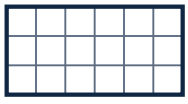


6) Area = 16 square units

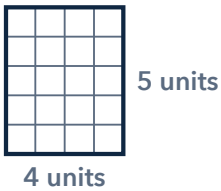


Part 2 Rows times columns Multiply instead of counting.

7) 3 × 6 = 18 square units



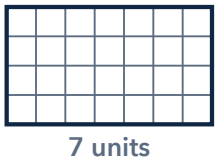
8) 5 × 4 = 20 square units



Area: Practice (continued)

Part 2 Rows times columns (continued)

9) 4 × 7 = 28 square units



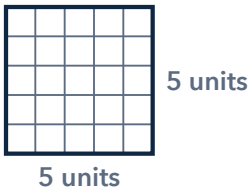
10) 6 × 3 = 18 square units



11) 2 × 9 = 18 square units



12) 5 × 5 = 25 square units



Part 3 Use the side lengths No squares drawn this time. Multiply the sides.

13) 5 × 8 = 40 square cm



14) 7 × 4 = 28 square in



15) 6 × 9 = 54 square m



16) 3 × 8 = 24 square ft



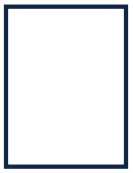
Area: Practice (continued)

Part 3 Use the side lengths (continued)

17) 9 × 7 = 63 square cm

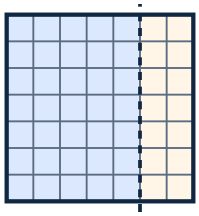


18) 8 × 6 = 48 square in



Part 4 Break it apart Find the area of each part. Add them.

19) 7 × 7 = 49

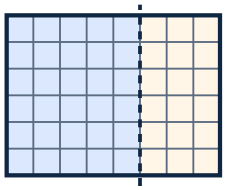


7 × 5 = 35

7 × 2 = 14

35 + 14 = 49

20) 6 × 8 = 48

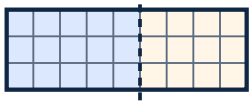


6 × 5 = 30

6 × 3 = 18

30 + 18 = 48

21) 3 × 9 = 27



3 × 5 = 15

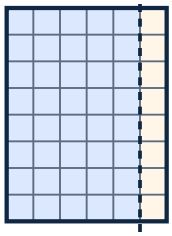
3 × 4 = 12

15 + 12 = 27

Area: Practice (continued)

Part 4 Break it apart (continued)

22) $8 \times 6 =$ 48



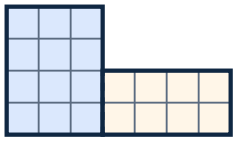
$8 \times 5 =$ 40

$8 \times 1 =$ 8

40 + 8 = 48

Part 5 Two rectangles Find each area. Add them.

23)

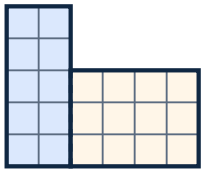


4 × 3 = 12

2 × 4 = 8

12 + 8 = 20

24)

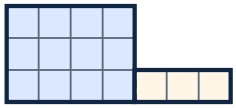


5 × 2 = 10

3 × 4 = 12

10 + 12 = 22

25)



3 × 4 = 12

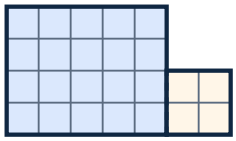
1 × 3 = 3

12 + 3 = 15

Area: Practice (continued)

Part 5 Two rectangles (continued)

26)



4 × 5 = 20

2 × 2 = 4

20 + 4 = 24

Part 6 Rectangles No help this time. Multiply the sides.

27) 6 cm by 7 cm: area = 42 sq cm

28) 8 cm by 3 cm: area = 24 sq cm

29) 9 cm by 4 cm: area = 36 sq cm

30) 5 cm by 6 cm: area = 30 sq cm

31) 7 cm by 8 cm: area = 56 sq cm

32) 3 cm by 9 cm: area = 27 sq cm

33) 5 cm by 9 cm: area = 45 sq cm

34) 2 cm by 8 cm: area = 16 sq cm

35) 4 cm by 8 cm: area = 32 sq cm

36) 9 cm by 8 cm: area = 72 sq cm

Part 7 Squares No help this time. All four sides are the same.

37) 6 m by 6 m square: area = 36 sq m

38) 3 in by 3 in square: area = 9 sq in

39) 8 cm by 8 cm square: area = 64 sq cm

40) 4 ft by 4 ft square: area = 16 sq ft

41) 9 m by 9 m square: area = 81 sq m

42) 7 in by 7 in square: area = 49 sq in

Area: Practice (continued)

Part 8 Find the missing side No help this time. Side × side = area. What is the other side?

43) 4 cm × 6 cm = 24 sq cm

44) 5 cm × 7 cm = 35 sq cm

45) 3 cm × 9 cm = 27 sq cm

46) 6 cm × 8 cm = 48 sq cm

47) 7 cm × 4 cm = 28 sq cm

48) 9 cm × 5 cm = 45 sq cm

49) 2 cm × 8 cm = 16 sq cm

50) 6 cm × 7 cm = 42 sq cm

Part 9 Remember? Problems from earlier packets.

51) 8 × 60 = 480

52) Round 347 to the nearest hundred: 300

53) 9 groups of 4 = 36

54) 6 sixths = 1 whole

55) 618 − 349 = 269

56) 2 fourths = 1 half

Part 10 Word problems Draw the rectangle if it helps.

57) A rug is 7 feet long and 3 feet wide. What is its area?

Number sentence 7 × 3 = 21

Answer 21 square feet

Draw or show your work

Area: Practice (continued)

Part 10 Word problems (continued)

58) A patio has 8 rows of square tiles with 4 tiles in each row. How many tiles cover it?

Number sentence $8 \times 4 = 32$

Answer 32 tiles

Draw or show your work

59) A poster is 9 inches long and 6 inches wide. What is its area?

Number sentence $9 \times 6 = 54$

Answer 54 square inches

Draw or show your work

60) A room is 10 feet long and 9 feet wide. What is its area?

Number sentence $10 \times 9 = 90$

Answer 90 square feet

Draw or show your work

61) Challenge: A room is 8 feet by 6 feet. A rug covers 20 square feet. How much floor is not covered?

Number sentence $8 \times 6 = 48,\ 48 - 20 = 28$

Answer 28 square feet

Draw or show your work

Check-In: Area

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Rows times columns

1) 2 × 5 = 10 square units

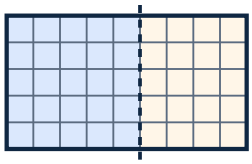


2) 3 × 3 = 9 square units



Part 2 Break it apart

3) 5 × 9 = 45



5 × 5 = 25

5 × 4 = 20

25 + 20 = 45

Part 3 Area

4) 9 ft by 4 ft: area = 36 sq ft

5) 6 in by 8 in: area = 48 sq in

6) 9 in by 8 in: area = 72 sq in

7) 6 in by 7 in: area = 42 sq in

Part 4 Find the missing side

8) 5 cm × 5 cm = 25 sq cm

9) 3 cm × 7 cm = 21 sq cm